

Week 2

TEAM MANUS

Progress

- ▶ Took inventory of items in our lockers
- ▶ Created a group chat
- ▶ Brainstormed design ideas

Issues

- ▶ The arm uses rubber bands
- ▶ The wiring is messy
- ▶ There isn't a claw
- ▶ We must shift center of gravity lower to prevent falling

Goals

- ▶ Find a way to increase resistance of the arm so that it can stay up at any angle
- ▶ Clean up the wiring
- ▶ Build the claw

Week 3

TEAM MANUS

Progress

- ▶ Held a productive Team meeting
- ▶ Established project timeline & goals
 - Fix the circuitry and design housing
 - Make the arm stable & presentable
 - Add a claw to the end
- ▶ Divide & Conquer
 - Made two teams
 - Will work simultaneously
 - Assigned different tasks to all team members

MANUS

Purdue ECE department
Design Lead: Carlos Lopez, Project Manager: Amitha
Thomas, Project Partner Liason: Abdullah Husain
Siddiqui

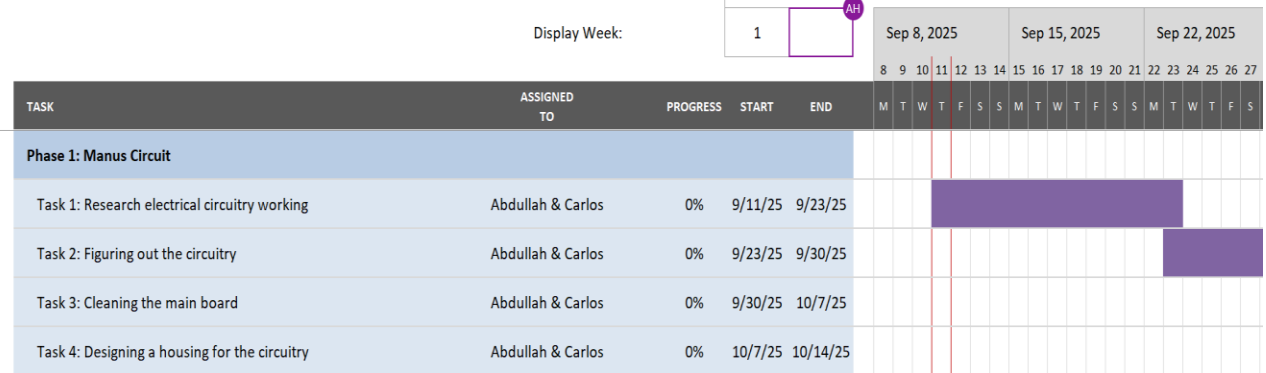
Project Start:

Thu, 9/11/2025

Display Week:

1

SIMPLE GANTT CHART by Vertex42.com
<https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.ht>



Issues

- ▶ Lack of insights into the project
- ▶ Understanding the working of the circuitry
- ▶ Understanding the “why” behind the arm falling

Goals

- ▶ Research ways to increase resistance in the joints of the arms
 - ▶ Study and research about mechanical joints
- ▶ Research working of electrical circuits
 - ▶ Demonstrate the ability to recognize different wires in the circuitry
- ▶ Study electrical and mechanical aspects of the project

Week 5

TEAM MANUS

Progress

- ▶ Reached out to an ex-Manus team member to arrange a meeting and gain insight
 - ▶ We agreed to meet today during EPICS
- ▶ Decided meeting timeline with Dr King. (Team prefers to meet on October 7th, but his availability will be prioritized)

Issues

- ▶ Time conflicts with midterms and career fair
- ▶ Can't work on the claw until we meet with Dr.King
 - ▶ Is it for a game or just for display?
- ▶ Need to come up with a budget.
 - ▶ We don't know what materials we need since we don't know what works and what doesn't

Goals

- ▶ Ask productive questions to Dr.King and the old team member to move forward with the project.
 - ▶ What do you want to accomplish with the claw?
 - ▶ Why is the arm metal?
 - ▶ How does the circuitry work?
 - ▶ How to connect the power supply and make the arm operate?

Week 6

TEAM MANUS

Progress

- ▶ Met with the previous members of the project and asked questions
 - ▶ Figured out that we should move the arm back
 - ▶ Make the base taller
 - ▶ Drill holes in the arm to reduce weight
 - ▶ Increase gear ratio
- ▶ Made the team budget
 - ▶ Total came up to 295 dollars

Issues

- ▶ Once we take apart the base, we can't put it back together again due to how it was built.
 - ▶ Need to start designing a taller base
- ▶ We need to increase the gear ratio to help make the arm more stable.
- ▶ Finalize the circuitry by soldering it.
 - ▶ Figure out the circuitry and how to plug the arm in

Goals

- ▶ Meet with a member of the previous team who did the circuitry
- ▶ Start designing the base
- ▶ Work on the design document and presentation
- ▶ Start designing the box in which the circuitry would go in
- ▶ Create better documentation

Week 8

TEAM MANUS

Progress

- ▶ Presented Design Document and received feedback
- ▶ Met with previous Manus Member, Owen.
 - ▶ Emphasized the importance of research
 - ▶ Showed us how to operate the arm.
- ▶ Decided to complete more research and split up respectively

Issues

- ▶ When operating, the arm did tend to fall
- ▶ Rubber bands are not effective and start to deteriorate after a long time.
- ▶ Identify how much power is needed to power arm.

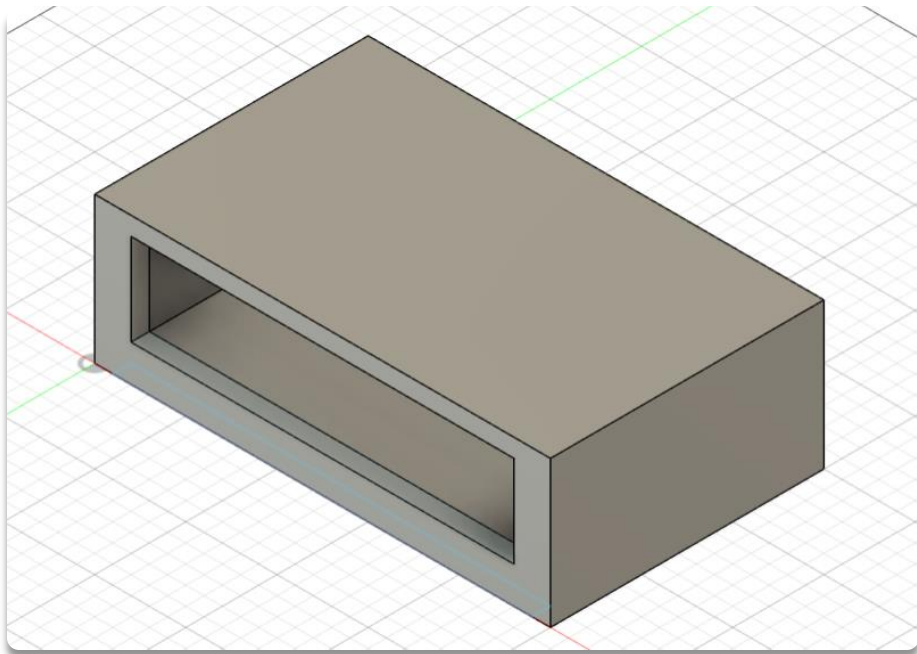
Goals

- ▶ Come up with solutions, backed up with evidence and links, and present to team
 - Motors
 - Other resistance methods
 - Arduino
 - Drivers

Week 9

TEAM MANUS

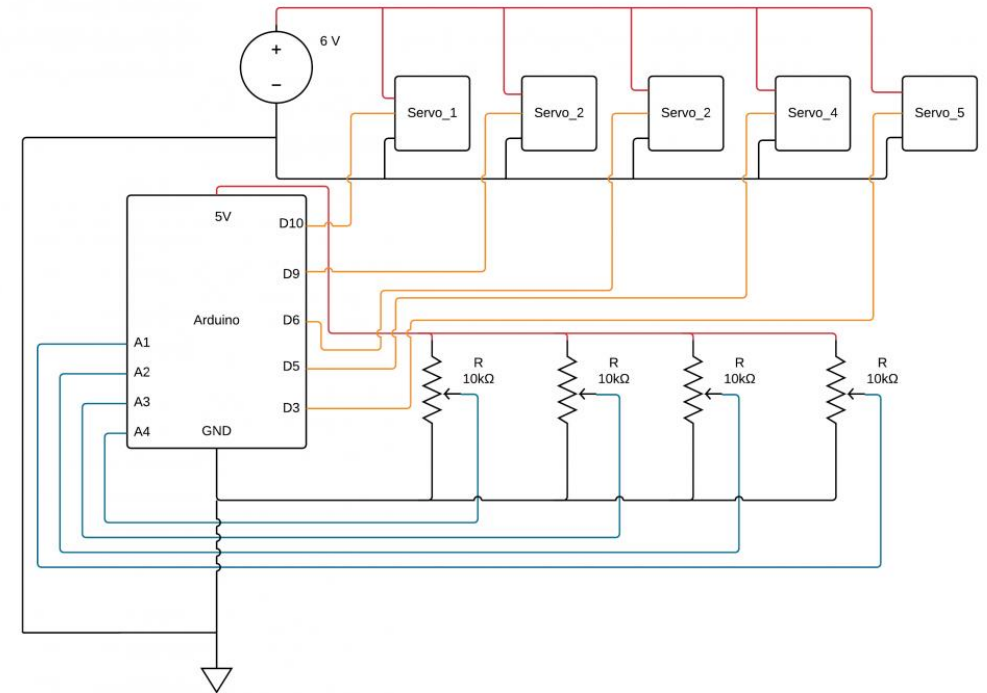
Progress



- ▶ Presented Design Document and received feedback from Dr. King
- ▶ Met with Dr King
 - ▶ Emphasized the importance of research
 - ▶ Showed him our ideas and plans for the project.
- ▶ Decided to complete more research and split up respectively
- ▶ Designed a prototype of the housing for the circuitry

Issues

- ▶ When operating, the arm did tend to fall
- ▶ Rubber bands are not effective and start to deteriorate after a long time.
- ▶ Identify how much power is needed to power arm and the 4 fans we plan to attach for cooling
- ▶ Identify electrical variables accurately
- ▶ Make a circuit diagram
- ▶ Identify strength of current rods vs replacement (3d print filament), determine a sufficient replacement



Example

Goals

- ▶ Preparing a circuit diagram with accurate values for resistance, voltage, current, etc.
- ▶ Coming up with solutions, backed up with evidence and links, and present to team
 - Motors
 - Other resistance methods
 - Arduino
 - Cooling Fans

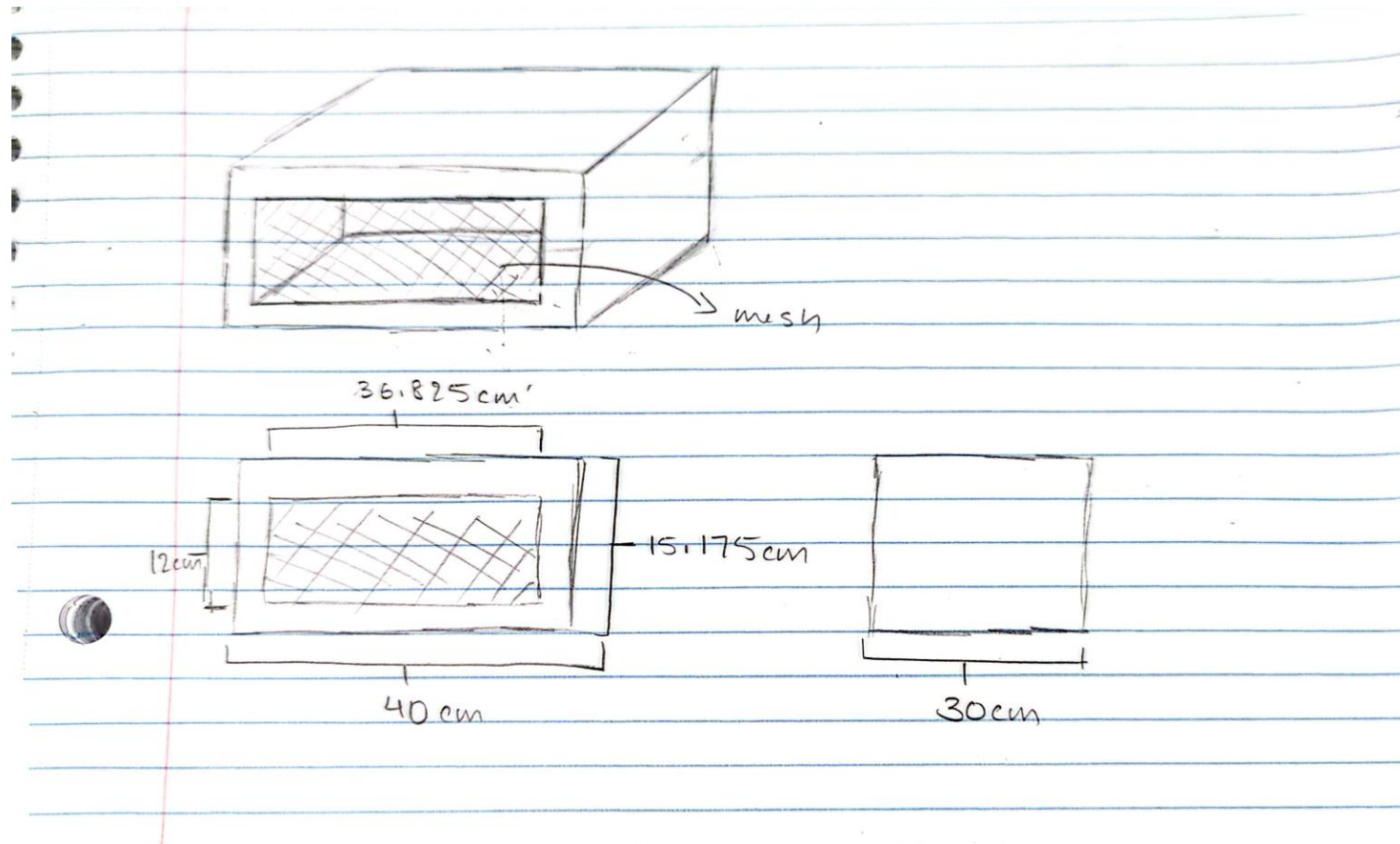
Week 10

TEAM MANUS

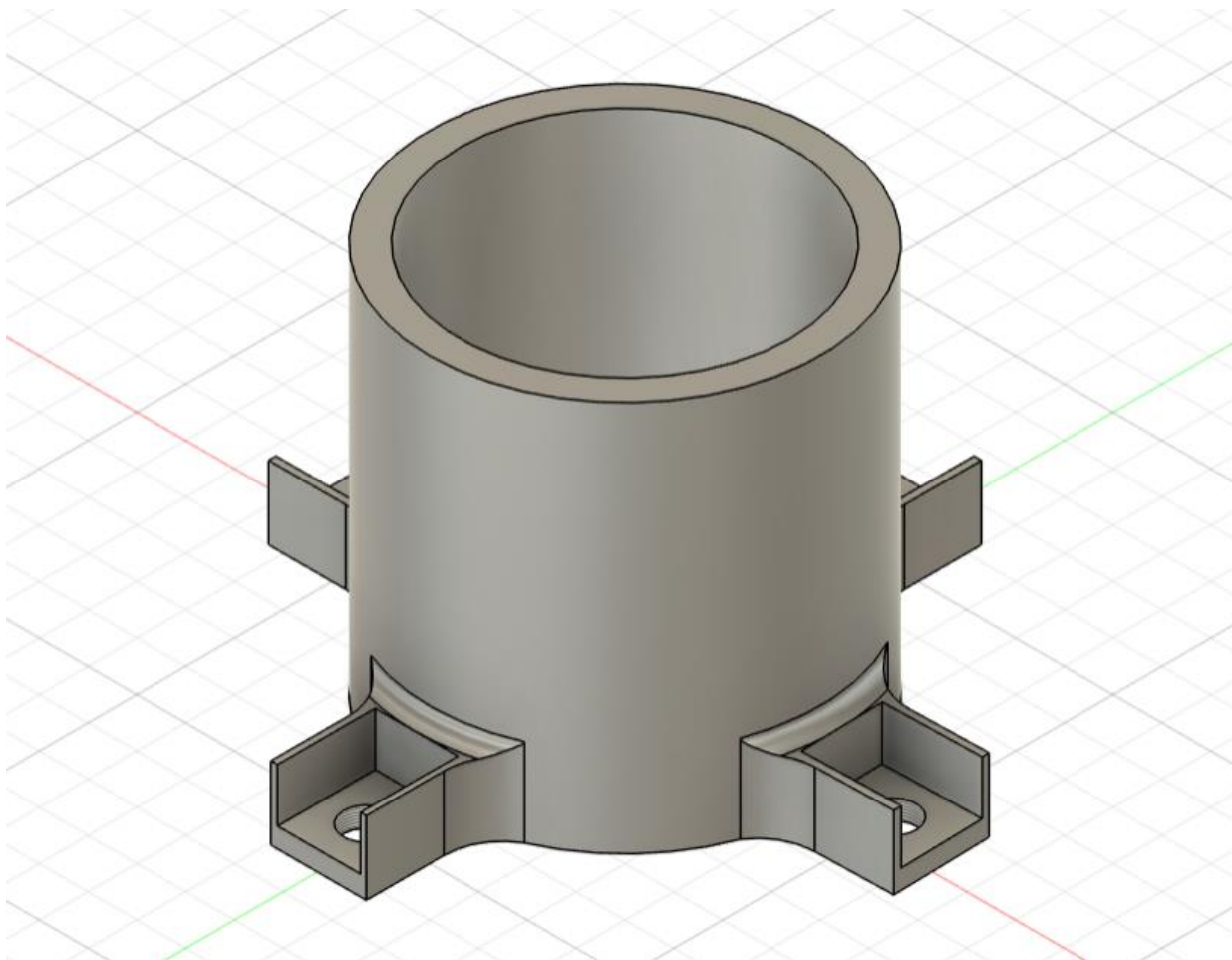
Progress

- ▶ Met to discuss how to move forward
- ▶ Measured the length and weight of the arms to find the correct motor
- ▶ Successfully filed a purchase request for new drivers and motors
- ▶ Prepared for cutting the wooden housing by doing a dimensional analysis
- ▶ Sketched the wooden housing, found what pieces we must build for the assembly
- ▶ Decided to power the fans directly instead of through a power supply
- ▶ Designed a taller base for the arm





- We need to cut a 40x30cm base, 2 x (30x12)cm side pieces of 5/8" plywood
- 3D print 2x rectangle frames to sandwich mesh between, (40x15.175)cm



Issues

- ▶ Cutting the wooden housing
- ▶ Integrating mesh vent into the housing
- ▶ 3-D printing the new base with the new motor

Goals

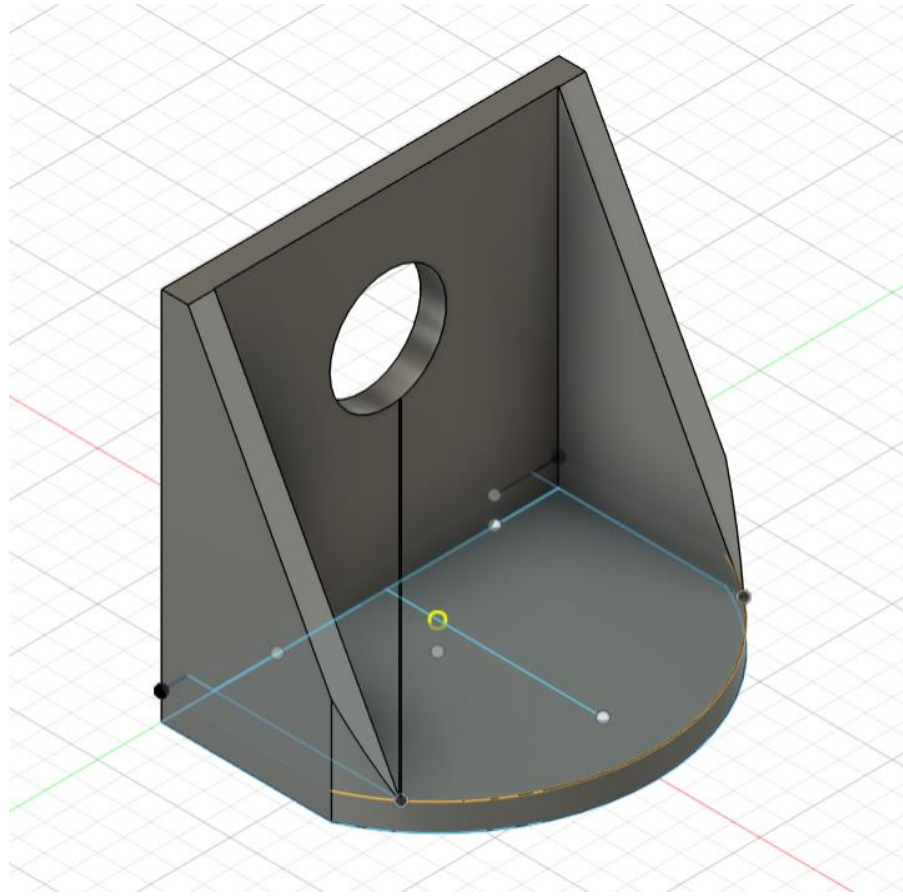
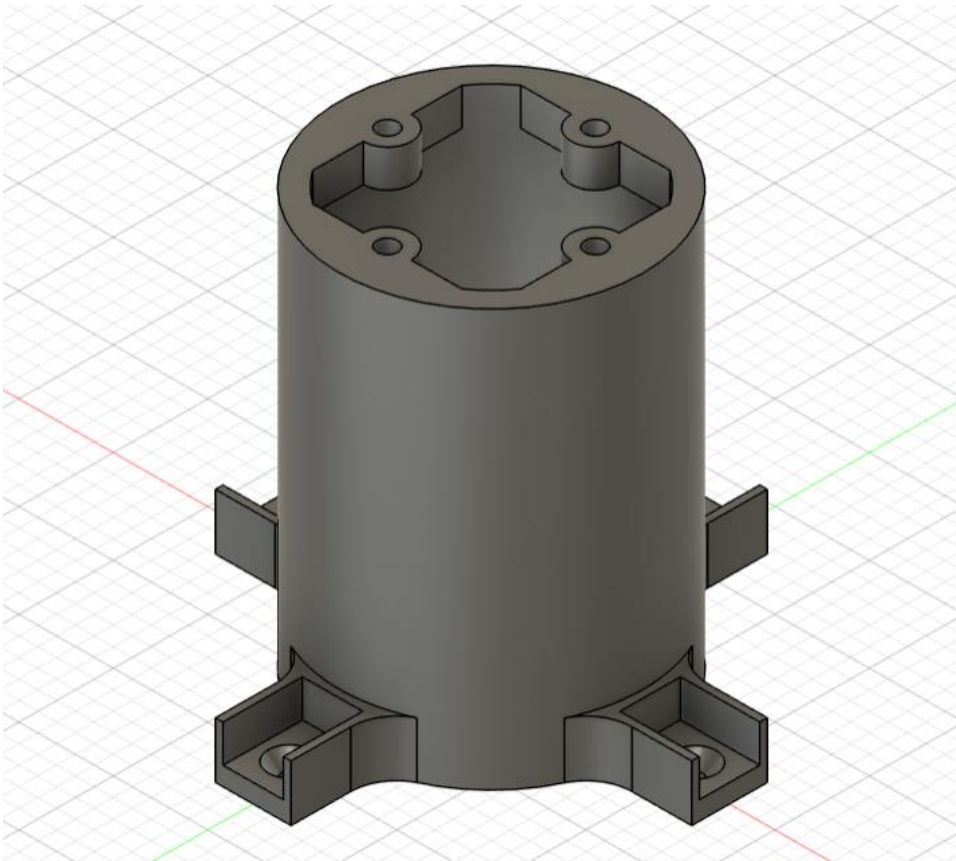
- ▶ Cut out the wooden housing
- ▶ 3-D Printing the blueprint for the new base
- ▶ Research:
 - ▶ 3-D printing (Amitha & Gordon)
 - ▶ Arduino (Abdullah & Carlos)

Week 11

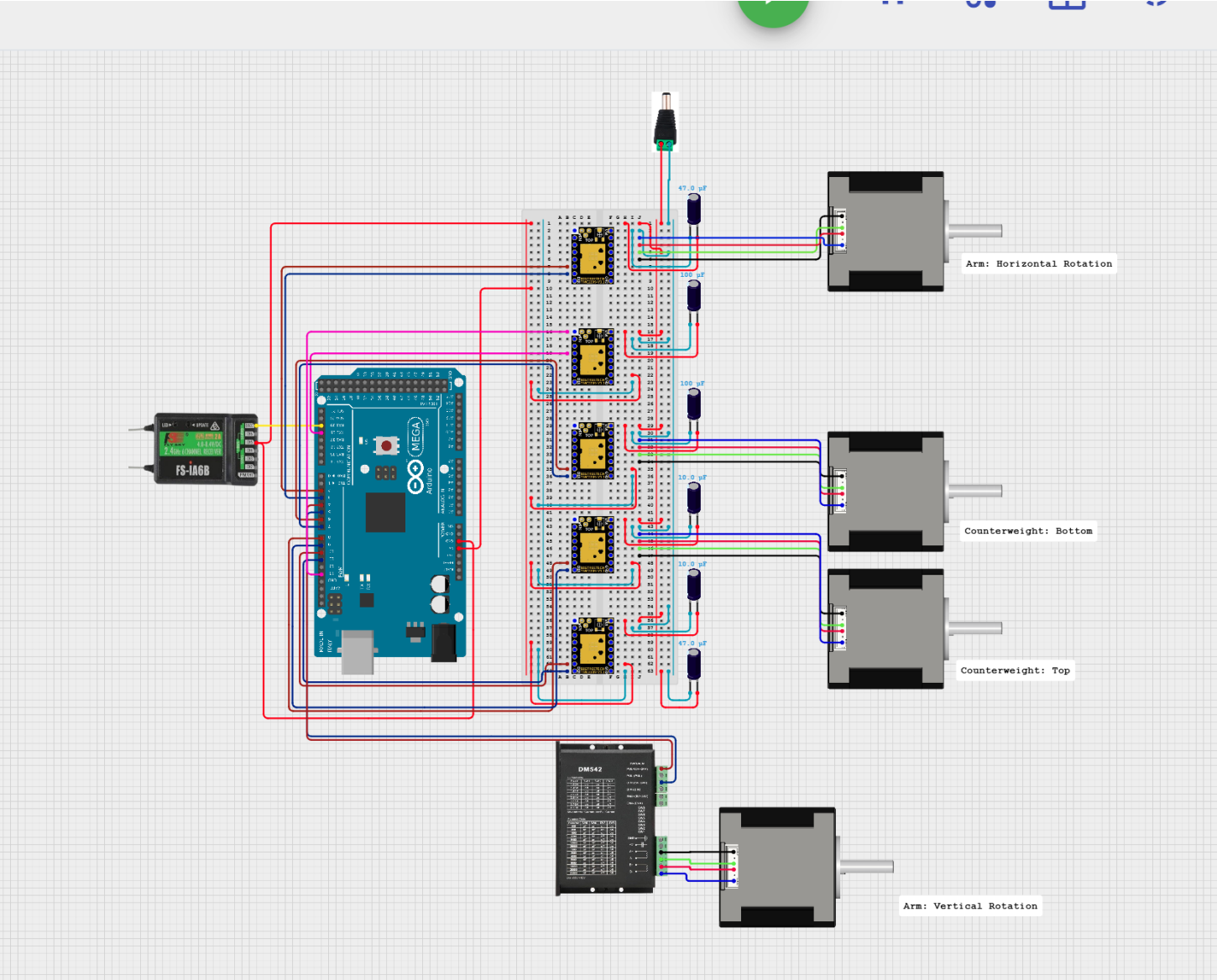
TEAM MANUS

Progress

- ▶ Housing is glued
- ▶ Wiring diagram is complete for the new motor and driver
- ▶ Prototype of our new base has been printed
- ▶ Fixed issues with the initial base and modeled a better version
- ▶ Modeled a new shoulder bracket that can fit the Nema 23 stepper motor







Issues

- ▶ Problem with how to attach fans and where to place them
- ▶ Need to print out the shoulder bracket to make sure it works and fits with the motor

Goals

- ▶ Finish taking apart old motor bracket
- ▶ Implement files into our GitHub
- ▶ Finish 3D model of complete housing
- ▶ Research:
 - ▶ 3-D printing (Amitha & Gordon)
 - ▶ Arduino (Abdullah & Carlos)

Week 12

TEAM MANUS

Progress

- ▶ Organized GitHub
- ▶ 3D printed two parts that were designed previously
- ▶ Adjusted the CAD of the shoulder bracket

 **ECE-Outreach-Manus** Public

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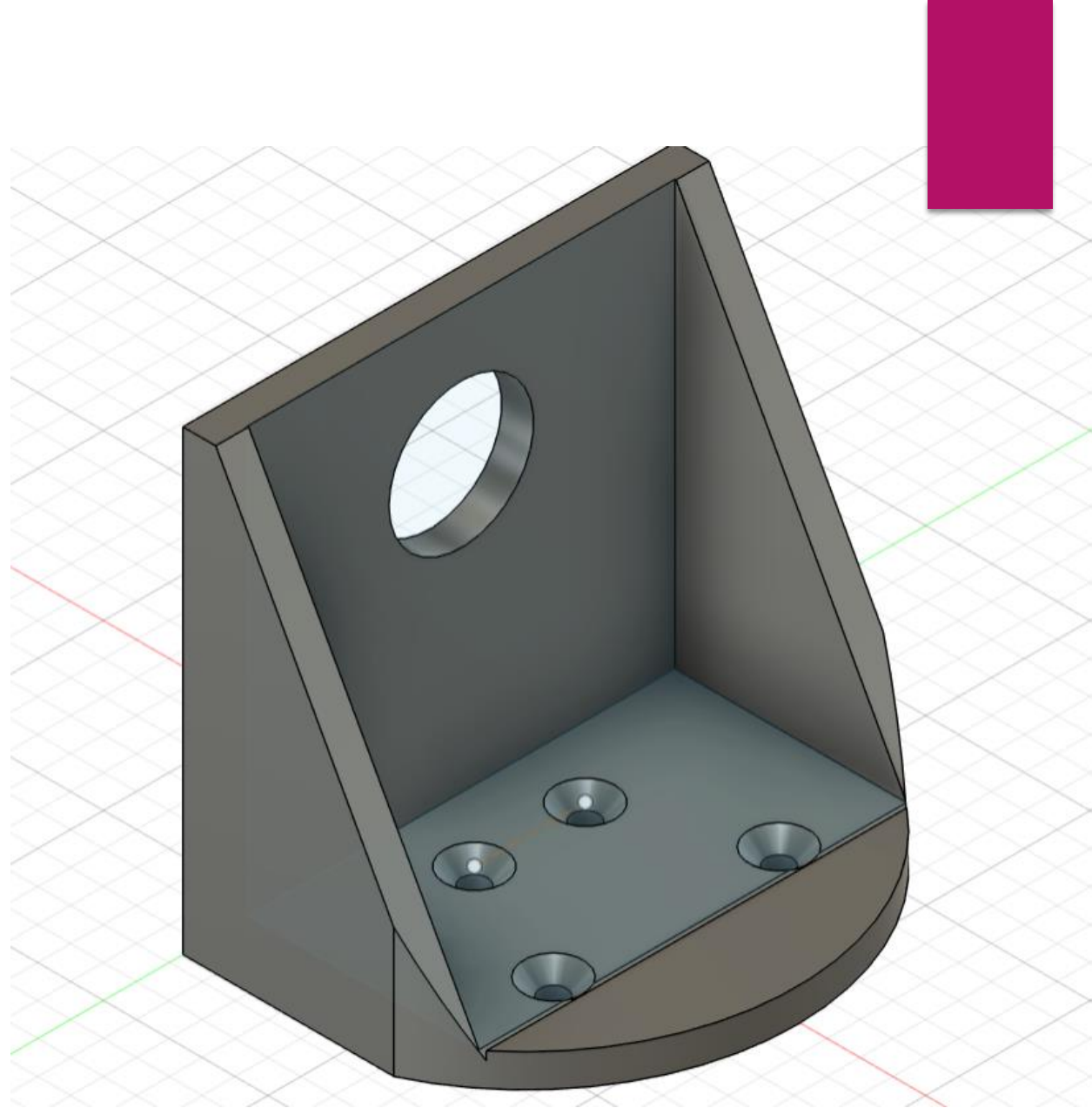
 main  1 Branch  0 Tags

   Code 

 **CLOPZ957** Create Videos ab6f320 · 1 minute ago  12 Commits

 Additional Resources	Add locker combo information to Info.md	6 minutes ago
 Arduino Code	Create Project_Manus.ino	last month
 Assembly Steps (Previous)	Update README.md	1 hour ago
 CAD Files	Add README for CAD Files directory	5 minutes ago
 Documents	Create Docs	4 minutes ago
 Helpful Materials	Create Videos	1 minute ago

 README



Issues

- ▶ Items have still not been delivered yet
- ▶ Assembly before the event to make the arm fit for display

Goals

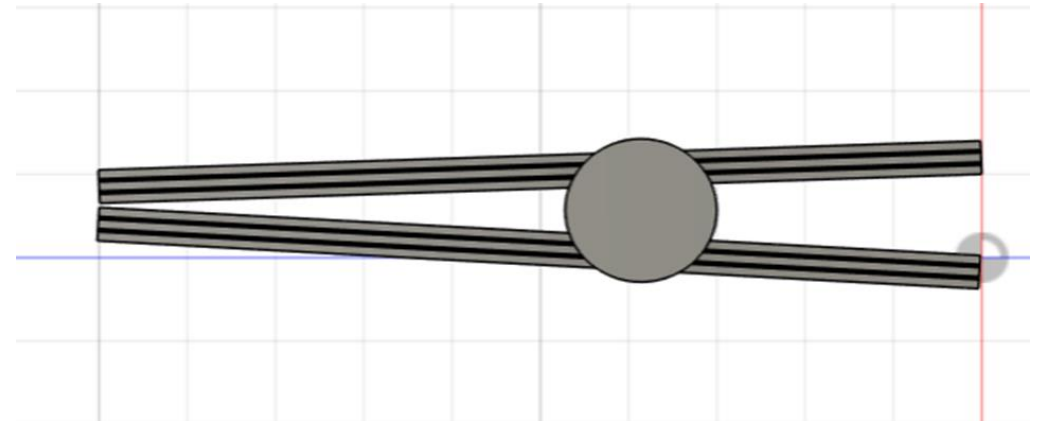
- ▶ Fit the new, more powerful motor – Carlos/Abdullah
- ▶ Reassemble the gears – Abdullah/Gordon
- ▶ CAD for mesh frame – Amitha/Gordon
- ▶ Continue Github
- ▶ Final assembly to make the arm functional – Team MANUS

Week 13

TEAM MANUS

Progress

- ▶ Organized and finalized the poster
- ▶ 3D modeled the arm (with longer shafts)
- ▶ 3D printed the new base and shoulder cage
- ▶ Prepared arm components for final assembly



Issues

- ▶ The circular disc for one of the gears broke
- ▶ Receive the new 3-D printed disc



Goals

- ▶ Prepare for the presentation
- ▶ Meet next week to assemble the arm

Week 14

TEAM MANUS

Progress

- ▶ Submitted the poster
- ▶ Discussed assembling the arm
- ▶ The 3D printed parts have been printed, ready for assembly

Issues

- ▶ We still haven't received the parts we ordered
- ▶ We need to reassemble the arm in its original configuration to be ready in time for the presentation.

Goals

- ▶ Prepare for the presentation Thursday
- ▶ Finalize details to make the arm more easily disassembled